SUMMER INTERNSHIP REPORT

on

Design and Development of "ReNest": A Full Stack Student Marketplace Web Application

Submitted in the Partial Fulfilment of the Requirement for the Award of

Bachelor of Technology

in

Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

Submitted By

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(Recognized by AICTE, Govt. of India & Affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, UP)

AKTU College Code -361

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CERTIFICATE

This is to certify that the Summer Internship Report entitled "Design and Development of ReNest: A Full Stack Student Marketplace Web Application" **submitted by** Mr. Utkarsh Yadav **(Roll No.** 2403611530068**), a student of** Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Machine Learning) **at** R.R. Institute of Modern Technology, Lucknow**, affiliated to** Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow**, is a bonafide record of the work carried out by him during the** 40-day Summer Internship **under my guidance and supervision.**

The work presented in this report is original and has been completed as a part of the academic requirements for the award of the degree of Bachelor of Technology**. To the best of my knowledge and belief, this report has not been submitted, either wholly or partially, to any other university or institution for the award of any degree, diploma, or other academic qualification.**

During the internship, the student demonstrated dedication, sincerity, and a strong willingness to learn modern web development technologies. The project involved the design and implementation of a full-stack web application using technologies such as Node.js, Express.js, MongoDB, HTML, CSS, JavaScript, JWT Authentication, and RESTful APIs**. The student successfully applied theoretical concepts to solve practical problems and completed the assigned work in a systematic and professional manner.**

I wish him success in all his future academic and professional endeavors.

Date: ___________________

Place: Lucknow

Faculty Guide	Head of Department
Mr. Bhavesh Mathur	Mr. Chandan Kr. Gupta
Department of Computer Science & Engineering (AI & ML)	**Department of Computer Science & Engineering (AI & ML)**
R.R. Institute of Modern Technology	**R.R. Institute of Modern Technology**
Signature	Signature & Seal

DECLARATION

I, Utkarsh Yadav**, Roll No.** 2403611530068**, a student of** Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Machine Learning) **at** R.R. Institute of Modern Technology, Lucknow**, hereby declare that the Summer Internship Report entitled** "Design and Development of ReNest: A Full Stack Student Marketplace Web Application" **is an original record of the work carried out by me during my** 40-day Summer Internship **under the guidance of** Mr. Bhavesh Mathur**, Faculty Guide, Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning.**

I further declare that this project has been developed solely for the purpose of partial fulfilment of the requirements for the award of the Bachelor of Technology **degree under** Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow**.**

I certify that the work presented in this report is original and has not been submitted, either wholly or partially, to any other university, institution, or organization for the award of any degree, diploma, certificate, or any other academic qualification.

The information, observations, analyses, and conclusions presented in this report are based on my own learning, practical implementation, and internship experience. Wherever the work or ideas of others have been referred to, appropriate acknowledgment has been provided in the references section.

I take full responsibility for the authenticity and accuracy of the contents of this report.

Place: **Lucknow**

Date: ____________________

Signature of the Student

(Utkarsh Yadav)

Roll No. 2403611530068

ACKNOWLEDGEMENT

The successful completion of this Summer Internship and the development of the ReNest – A Full Stack Student Marketplace Web Application **would not have been possible without the guidance, encouragement, and support of several individuals and organizations.**

First and foremost, I express my sincere gratitude to Mr. Bhavesh Mathur**, Faculty Guide, Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning),** R.R. Institute of Modern Technology, Lucknow**, for his valuable guidance, continuous motivation, constructive suggestions, and constant support throughout the internship period. His technical insights and encouragement helped me successfully complete this project.**

I would also like to express my heartfelt thanks to the Department of Computer Science & Engineering **and the management of** R.R. Institute of Modern Technology, Lucknow**, for providing me with the opportunity to undergo this Summer Internship as a part of my academic curriculum. The knowledge, resources, and learning environment provided by the institution played a significant role in enhancing my technical and professional skills.**

I am thankful to the training organization and all the trainers **for sharing their valuable knowledge of Full Stack Web Development and for providing practical exposure to modern software development technologies. The training sessions helped me strengthen my understanding of frontend development, backend development, database management, RESTful APIs, version control, testing methodologies, and professional software development practices.**

I would also like to thank my classmates and friends for their cooperation, constructive discussions, and encouragement during the internship. Their support helped me overcome several technical challenges encountered during the development of the project.

Finally, I express my deepest gratitude to my parents and family members for their unconditional love, continuous encouragement, patience, and unwavering belief in me throughout my academic journey. Their constant support has been the greatest source of motivation behind the successful completion of this internship and the preparation of this report.

I sincerely acknowledge everyone who, directly or indirectly, contributed to the successful completion of this internship and this project.

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ABSTRACT

The rapid growth of digital technology has transformed the way people buy and sell products, making online marketplaces an integral part of everyday life. However, most existing e-commerce platforms are designed for the general public and often lack features that address the specific needs of college students. Students frequently possess textbooks, electronic gadgets, stationery, furniture, and other reusable items that become unnecessary after a semester or an academic year. Finding trustworthy buyers or sellers within the college community can be difficult, resulting in unused resources and unnecessary expenditure. To address this problem, ReNest was conceptualized and developed as a dedicated student marketplace that promotes the reuse of products within a secure campus environment.

ReNest is a Full Stack Web Application designed to enable college students to buy, sell, and manage second-hand products conveniently through a user-friendly web interface. The application provides a centralized platform where authenticated users can register, log in securely, create product listings, browse available products, view detailed product information, manage their own listings, and contact sellers for further communication. By restricting the platform to the student community, the system aims to improve trust, encourage sustainable reuse of resources, and reduce unnecessary spending.

The project has been developed using modern web technologies including HTML5, CSS3, and JavaScript for the frontend, while Node.js and Express.js are used to implement the backend services. MongoDB, along with Mongoose, is employed for data storage and database management. Secure user authentication is implemented using JSON Web Tokens (JWT), while bcrypt is used to encrypt user passwords before storing them in the database. Additional development tools such as Postman, Git, and GitHub were utilized for API testing, version control, and collaborative development practices.

The system follows a modular architecture consisting of authentication, product management, user management, database management, and REST API modules. This architecture improves maintainability, scalability, and ease of future enhancements. Proper

The development of ReNest provided valuable practical exposure to the complete Software Development Life Cycle (SDLC), including requirement analysis, system design, implementation, testing, debugging, and documentation. Throughout the internship, significant knowledge was gained in backend development, RESTful API design, database modeling, authentication mechanisms, project organization, and version control workflows. The project also strengthened problem-solving abilities and enhanced understanding of how modern full-stack applications are designed and deployed in real-world environments.

Overall, ReNest demonstrates the practical application of full-stack web development technologies in solving a real-world problem faced by college students. The project successfully achieves its objective of providing a secure, user-friendly, and scalable platform for buying and selling second-hand products within a campus community while establishing a strong foundation for future enhancements such as real-time messaging, online payments, product reviews, notifications, and mobile responsiveness.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

The advancement of information technology has transformed the way people buy, sell, and exchange goods. Online marketplaces have become one of the most popular digital platforms because they provide convenience, accessibility, and faster communication between buyers and sellers.

College students often need books, electronic gadgets, furniture, stationery, and other academic essentials. After completing a semester, many of these items remain unused, while other students require them. However, there is no dedicated platform within most colleges that allows students to exchange these products safely and efficiently.

ReNest **is a Full Stack Student Marketplace Web Application developed to solve this problem. It provides a secure platform where students can register, browse products, list items for sale, manage their listings, and contact sellers within their campus community.**

The application is developed using HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Mongoose, JWT Authentication, and REST APIs**. It follows a modular architecture that ensures scalability, maintainability, and secure user authentication.**

Key Highlights of ReNest

- **Student-focused online marketplace**
- **Secure user authentication using JWT**
- **Product listing and management**
- **Browse available products**
- **User profile management**
- **RESTful API architecture**
- **MongoDB database integration**
- **Responsive and user-friendly interface**

1.2 Background of the Project

The increasing use of digital platforms has changed the way people buy and sell products. Online marketplaces such as Amazon, Flipkart, and OLX have simplified product transactions for the general public. However, these platforms are not specifically designed for college students and often lack a trusted campus-based environment.

Students frequently own books, calculators, laptops, bicycles, furniture, and other items that become unnecessary after a semester. At the same time, new students often need these products but struggle to find affordable options within their college.

ReNest **was developed to bridge this gap by providing a dedicated marketplace exclusively for students. The platform promotes the reuse of products, reduces unnecessary expenses, and creates a trusted environment for buying and selling within the college community.**

1.3 Need of the Project

The development of ReNest is motivated by the following needs:

- **To provide a dedicated marketplace for college students.**
- **To simplify buying and selling of second-hand products.**
- **To reduce dependency on social media groups and informal communication.**
- **To promote the reuse of educational resources and reduce waste.**
- **To provide secure authentication for users.**
- **To maintain organized product information.**
- **To improve accessibility through a web-based platform.**
- **To create a trusted campus community for transactions.**

1.4 Objectives of the Project

The main objectives of the project are:

- **To develop a secure student marketplace.**
- **To implement user authentication using JWT.**
- **To allow students to register and manage their accounts.**
- **To enable users to add, edit, and delete product listings.**
- **To provide product browsing and detailed product information.**
- **To implement RESTful APIs for communication between frontend and backend.**

- **To securely store user and product information using MongoDB.**
- **To gain practical experience in Full Stack Web Development.**
- **To understand real-world software development practices.**

1.5 Scope of the Project

The scope of ReNest includes the following functionalities:

Current Scope

- **User Registration and Login**
- **Secure Authentication**
- **User Profile Management**
- **Product Listing**
- **Product Editing**
- **Product Deletion**
- **Marketplace Browsing**
- **Product Details Page**
- **Product Categories**
- **MongoDB Database Integration**
- **REST API Implementation**
- **Input Validation**
- **Error Handling**

Future Scope

- **Real-time Chat System**
- **Online Payment Gateway**
- **Wishlist Feature**
- **Product Search and Filters**
- **Ratings and Reviews**
- **College Email Verification**
- **Admin Dashboard**
- **Push Notifications**

- **Mobile Responsive Application**
- **Mobile App Development**

1.6 Mission

The mission of ReNest is to develop a secure, reliable, and user-friendly marketplace exclusively for college students that encourages the reuse of products and promotes sustainable resource sharing within educational institutions.

The project also aims to provide practical exposure to modern Full Stack Web Development technologies while implementing industry-standard software development practices.

Chapter 1 Summary

This chapter introduced the ReNest project by discussing its background, the need for a dedicated student marketplace, project objectives, scope, and mission. It also highlighted how the project addresses real-world challenges faced by students while providing an opportunity to apply modern web development technologies in a practical environment.

CHAPTER 2

ORGANIZATION / INTERNSHIP PROFILE

2.1 Organization Overview

The Summer Internship was undertaken as a part of the Bachelor of Technology curriculum at R.R. Institute of Modern Technology (RRIMT), Lucknow**, affiliated with** Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow**.**

The internship program was designed to provide practical exposure to modern software development technologies and bridge the gap between academic learning and industry practices. During the training, students were introduced to Full Stack Web Development concepts, project development methodologies, and software engineering principles through hands-on sessions and practical assignments.

Note: **Replace this section with the training organization's details once confirmed.**

2.2 Internship Details

Particular	Details
Internship Title	**Summer Internship on Full Stack Web Development**
Project Title	**ReNest – A Full Stack Student Marketplace Web Application**
Duration	**40 Days**
Student Name	**Utkarsh Yadav**
Roll Number	**2403611530068**
Branch	**Computer Science & Engineering (AI & ML)**
College	**R.R. Institute of Modern Technology, Lucknow**
University	**Dr. A.P.J. Abdul Kalam Technical University (AKTU)**
Faculty Guide	**Mr. Bhavesh Mathur**

2.3 Training Objectives

The primary objectives of the internship were:

- **To understand the fundamentals of Full Stack Web Development.**
- **To gain practical experience in frontend and backend technologies.**

- **To learn database design and management using MongoDB.**
- **To understand RESTful API development.**
- **To implement secure authentication mechanisms.**
- **To improve debugging and testing skills.**
- **To understand Git and GitHub workflow.**
- **To develop a real-world web application.**

2.4 Training Methodology

The internship followed a practical learning approach that combined theoretical concepts with hands-on implementation.

The methodology included:

- **Classroom-based technical sessions.**
- **Hands-on coding exercises.**
- **Mini practical assignments.**
- **Project-based learning.**
- **Regular debugging sessions.**
- **API testing using Postman.**
- **Version control using Git and GitHub.**
- **Continuous project evaluation and documentation.**

2.5 Technologies Covered During Training

Category	Technologies / Tools
Frontend	**HTML5, CSS3, JavaScript**
Backend	**Node.js, Express.js**
Database	**MongoDB, Mongoose**
Authentication	**JWT, bcrypt**
API Testing	**Postman**
Version Control	**Git, GitHub**
Development Tool	**Visual Studio Code**

Category	Technologies / Tools
Browser	**Google Chrome**

2.6 Learning Approach

During the internship, emphasis was placed on learning by implementation rather than theory alone. Each technology was introduced through practical demonstrations followed by coding exercises. The acquired knowledge was applied in developing the ReNest **project, enabling a better understanding of real-world software development practices.**

Chapter Summary

This chapter provided an overview of the internship, including the organization, objectives, training methodology, technologies covered, and the overall learning approach adopted during the 40-day internship program.

CHAPTER 3

INTERNSHIP OBJECTIVES

3.1 Internship Details

Particular	Details
Internship Title	**Summer Internship on Full Stack Web Development**
Project Title	**ReNest – A Full Stack Student Marketplace Web Application**
Duration	**40 Days**
Student Name	**Utkarsh Yadav**
Roll Number	**2403611530068**
Branch	**Computer Science & Engineering (Artificial Intelligence & Machine Learning)**
College	**R.R. Institute of Modern Technology, Lucknow**
University	**Dr. A.P.J. Abdul Kalam Technical University (AKTU)**
Faculty Guide	**Mr. Bhavesh Mathur**
Development Tools	**Visual Studio Code, Git, GitHub, Postman**
Database	**MongoDB**
Backend	**Node.js, Express.js**
Frontend	**HTML5, CSS3, JavaScript**

3.2 Project Overview

ReNest **is a Full Stack Student Marketplace Web Application developed to provide a secure platform where college students can buy and sell second-hand products within their campus community.**

The application enables authenticated users to:

- **Register and log in securely.**
- **Browse available products.**
- **Add new product listings.**

- **View product details.**
- **Edit or remove their own listings.**
- **Manage their user profile.**
- **Contact product sellers.**

The system is designed using a modular architecture, making it scalable, secure, and easy to maintain.

3.3 Objectives of the Internship

The internship was conducted with the following objectives:

- **To understand the complete Software Development Life Cycle (SDLC).**
- **To gain practical knowledge of Full Stack Web Development.**
- **To develop a real-world web application.**
- **To learn backend development using Node.js and Express.js.**
- **To understand MongoDB database design.**
- **To implement secure authentication using JWT.**
- **To design RESTful APIs.**
- **To improve debugging and testing skills.**
- **To understand project documentation.**
- **To apply classroom concepts to practical software development.**

3.4 Learning Objectives

During the internship, emphasis was placed on acquiring the following technical skills:

Technical Learning

- **HTML5 and Semantic Web Design**
- **CSS3 Styling and Responsive Design**
- **JavaScript Programming**
- **Node.js Development**
- **Express.js Framework**
- **MongoDB Database**
- **Mongoose ODM**
- **REST API Development**

- **JWT Authentication**
- **Password Hashing using bcrypt**
- **Git & GitHub Version Control**
- **API Testing using Postman**

3.5 Learning Outcomes

At the successful completion of the internship, the following outcomes were achieved.

Technical Skills

- **Full Stack Web Development**
- **Frontend Development**
- **Backend Development**
- **Database Design**
- **REST API Development**
- **User Authentication**
- **Secure Password Management**
- **CRUD Operations**
- **Debugging**
- **Software Testing**
- **Git Version Control**
- **Professional Documentation**

Professional Skills

- **Problem Solving**
- **Logical Thinking**
- **Time Management**
- **Team Collaboration**
- **Technical Communication**
- **Project Planning**
- **Documentation Skills**
- **Software Development Practices**

3.6 Key Features Developed During the Internship

Module	Features Implemented
Authentication	**Registration, Login, JWT Authentication, Password Encryption**
Product Management	**Add Product, Edit Product, Delete Product, View Product**
Marketplace	**Browse Products, Categories, Product Details**
User Module	**User Profile, My Listings**
Database	**MongoDB Collections, CRUD Operations**
Backend	**REST APIs, Middleware, Validation, Error Handling**

3.7 Internship Outcome

The internship provided practical exposure to modern software development by integrating frontend technologies, backend services, and database management into a single project.

The development of ReNest **enhanced my understanding of application architecture, secure authentication, RESTful API development, and project organization. It also strengthened my ability to analyze problems, design scalable solutions, and implement them using industry-standard technologies.**

Chapter Summary

This chapter presented the internship details, project overview, objectives, learning outcomes, and the major features implemented during the development of ReNest**. It also highlighted the technical and professional skills acquired throughout the internship.**

CHAPTER 4

TECHNOLOGIES USED

4.1 Introduction

The development of ReNest **involved the use of modern web development technologies to build a secure, scalable, and user-friendly student marketplace. Each technology was selected based on its functionality, performance, and suitability for full-stack application development.**

4.2 Technology Stack

Category	Technology	Purpose
Frontend	**HTML5**	**Structure of web pages**
Frontend	**CSS3**	**Styling and responsive design**
Frontend	**JavaScript**	**Client-side functionality and interactivity**
Backend	**Node.js**	**Server-side JavaScript runtime**
Backend	**Express.js**	**Web application framework**
Database	**MongoDB**	**NoSQL database for storing user and product data**
ODM	**Mongoose**	**Database modeling and schema management**
Authentication	**JWT**	**Secure user authentication**
Security	**bcrypt**	**Password hashing and encryption**
Configuration	**dotenv**	**Environment variable management**
API Testing	**Postman**	**Testing REST APIs**
Version Control	**Git**	**Source code management**
Repository	**GitHub**	**Project hosting and version control**
Development Tool	**Visual Studio Code**	**Code editor**
Browser	**Google Chrome**	**Testing and debugging**

4.3 Software Development Life Cycle (SDLC)

The development of ReNest followed the Software Development Life Cycle (SDLC), ensuring systematic planning, development, testing, and maintenance.

Phases of SDLC

1. **Requirement Analysis**
2. **Planning**
3. **System Design**
4. **Development**
5. **Testing**
6. **Deployment**
7. **Maintenance**

SDLC Workflow

Requirement Analysis

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Planning

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System Design

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Development

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Testing

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Deployment

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Maintenance

4.4 Technologies Description

HTML5

HTML5 was used to create the structure of all web pages including the Home page, Login page, Marketplace, Dashboard, Profile, and Product pages.

CSS3

CSS3 was used to design the user interface, implement responsive layouts, apply animations, and improve the overall appearance of the application.

JavaScript

JavaScript provides dynamic functionality such as form validation, API requests, DOM manipulation, event handling, and interaction with backend services.

Node.js

Node.js was used to execute JavaScript on the server and handle incoming client requests efficiently.

Express.js

Express.js simplified backend development by providing routing, middleware support, request handling, and REST API implementation.

MongoDB

MongoDB stores user accounts, product listings, authentication data, and other application information in a flexible NoSQL database.

Mongoose

Mongoose was used to define schemas, create models, and perform CRUD operations between the application and MongoDB.

JWT Authentication

JSON Web Tokens (JWT) provide secure authentication by generating tokens after successful login and verifying users before allowing access to protected routes.

bcrypt

bcrypt secures user passwords by converting them into hashed values before storing them in the database.

Postman

Postman was used to test REST APIs by sending HTTP requests and validating API responses during development.

Git & GitHub

Git was used to track project changes locally, while GitHub served as the remote repository for source code management and version control.

4.5 Advantages of the Technologies Used

- **Open-source technologies with strong community support.**
- **Fast and efficient backend using Node.js.**
- **Flexible NoSQL database with MongoDB.**
- **Secure authentication using JWT.**
- **Password protection through bcrypt hashing.**
- **Easy API development with Express.js.**
- **Efficient version control using Git and GitHub.**
- **Simple API testing using Postman.**
- **Modular architecture for future scalability.**

4.6 Limitations

- **JWT tokens require proper management for security.**
- **MongoDB requires careful schema planning to maintain consistency.**
- **Node.js is single-threaded, making CPU-intensive tasks less efficient.**
- **Internet connectivity is required to access the deployed application.**
- **Additional features such as real-time chat and payment integration are not implemented in the current version.**

4.7 Future Scope

The project can be enhanced with the following features:

- **Image upload using cloud storage.**
- **Real-time messaging between buyers and sellers.**
- **Wishlist functionality.**
- **Advanced product search and filters.**
- **College email verification.**
- **Product ratings and reviews.**
- **Admin dashboard.**
- **Push notifications.**
- **Online payment gateway integration.**
- **Responsive mobile application.**

Chapter Summary

This chapter presented the technologies, development tools, and software development methodology used during the implementation of ReNest**. It also discussed the advantages, limitations, and future enhancement possibilities of the project.**

CHAPTER 5
PROJECT DESCRIPTION

5.1 Project Introduction

ReNest is a Full Stack Student Marketplace Web Application developed to provide a secure platform where students can buy and sell second-hand products within their college community.

The application allows authenticated users to register, log in, browse products, add new product listings, manage their own products, and view detailed information about available items. By limiting the marketplace to students, the platform aims to build trust and encourage the reuse of useful products.

5.2 Problem Statement

Students often own items such as textbooks, electronic gadgets, calculators, furniture, bicycles, and stationery that are no longer required after a semester. Although these products are still useful, there is no dedicated platform within colleges where students can exchange them safely.

As a result, students usually depend on:

- WhatsApp groups
- Social media posts
- Personal contacts
- College notice boards

These methods are unorganized, difficult to manage, and often unreliable.

5.3 Proposed Solution

ReNest provides a centralized web platform where students can securely buy and sell second-hand products.

The system allows users to:

- Register and log in securely.
- Browse available products.
- Publish products for sale.
- Edit or remove their own listings.
- View product details.

- **Contact sellers.**
- **Manage their profile.**

This improves accessibility, transparency, and convenience while reducing unnecessary waste of usable products.

5.4 Project Goals

The major goals of ReNest are:

- **Develop a secure student marketplace.**
- **Simplify buying and selling within the campus.**
- **Promote the reuse of educational resources.**
- **Implement secure authentication using JWT.**
- **Provide an easy-to-use interface.**
- **Maintain organized product records.**
- **Develop scalable REST APIs.**
- **Gain practical experience in Full Stack Development.**

5.5 Major Modules

Module	Description
Authentication Module	**Handles user registration, login, JWT authentication, and logout.**
Product Module	**Manages product creation, editing, deletion, and viewing.**
User Module	**Handles user profile and product management.**
Database Module	**Stores user and product information using MongoDB.**
API Module	**Provides REST APIs for frontend and backend communication.**

5.6 Working of ReNest

The overall workflow of the application is as follows:

1. **User registers on the platform.**
2. **User logs in using registered credentials.**
3. **JWT token is generated after successful authentication.**
4. **User can browse available products.**
5. **Authenticated users can add new products.**
6. **Users can edit or delete their own listings.**
7. **Buyers can view product details and contact the seller.**

8. Product information is stored and retrieved from MongoDB through REST APIs.

5.7 System Architecture

The ReNest application follows a layered architecture consisting of four major layers.

Users

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Frontend (HTML, CSS, JavaScript)

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REST API Requests

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Backend (Node.js + Express.js)

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Authentication & Business Logic

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MongoDB Database (Mongoose)

5.8 Key Features

Authentication

- User Registration
- User Login
- JWT Authentication
- Password Encryption
- Protected Routes

Marketplace

- Browse Products
- Product Details
- Product Categories

- Product Description
- Product Price

User Features

- Add Product
- Edit Product
- Delete Product
- Manage Own Listings
- User Profile

Backend Features

- REST APIs
- MongoDB Integration
- Authentication Middleware
- Input Validation
- Error Handling

5.9 Benefits of the Project

- Dedicated marketplace for students.
- Secure user authentication.
- Easy product management.
- Organized database.
- Scalable architecture.
- User-friendly interface.
- Reduced dependency on informal selling methods.
- Promotes sustainable reuse of products.

Chapter Summary

This chapter described the ReNest project, its problem statement, proposed solution, objectives, working methodology, architecture, major modules, and key features. The project provides a secure and efficient marketplace that simplifies the buying and selling process within a college environment.

CHAPTER 6

EXISTING SYSTEM

6.1 Introduction

Before the development of ReNest, students generally relied on informal methods to buy and sell second-hand products. These methods lacked proper organization, security, and reliability, making the process inconvenient for both buyers and sellers.

6.2 Existing System

The commonly used methods for exchanging products among students include:

- WhatsApp Groups
- Telegram Groups
- Facebook Marketplace
- OLX
- Personal Contacts
- College Notice Boards

Although these platforms facilitate buying and selling, they are not specifically designed for college students and do not provide a trusted campus-based marketplace.

6.3 Limitations of the Existing System

The existing system suffers from several limitations:

- No dedicated marketplace for students.
- Difficult to verify genuine buyers and sellers.
- Product information is scattered across multiple platforms.
- No centralized product management.
- Lack of secure user authentication.
- Higher chances of fraudulent listings.
- No proper categorization of products.
- Difficult to manage old or sold listings.
- Communication depends on external applications.

6.4 Problems Identified

The major problems observed are:

Problem	Description
Lack of Trust	Users cannot easily verify whether the seller belongs to the same college.
Poor Organization	Products are shared randomly through messages or posts.
Difficult Search	Finding required products takes considerable time.
No User Management	No dedicated user profiles or product history.
No Listing Control	Sellers cannot efficiently manage their posted products.
Limited Security	Most platforms do not provide campus-specific authentication.

6.5 Drawbacks of the Existing System

- Time-consuming product search.
- Unorganized buying and selling process.
- Dependence on multiple communication platforms.
- Limited control over product listings.
- Higher possibility of fake or outdated advertisements.
- No dedicated platform for campus transactions.

CHAPTER 7
PROPOSED SYSTEM

7.1 Introduction

The proposed system, ReNest, is a Full Stack Student Marketplace Web Application developed to provide a secure and organized platform for buying and selling second-hand products within a college campus.

Unlike traditional methods such as social media groups or general online marketplaces, ReNest is specifically designed for students, providing a centralized platform with secure authentication, product management, and an easy-to-use interface.

7.2 Proposed System

ReNest provides the following functionalities:

- Secure user registration and login.
- JWT-based user authentication.
- Browse available products.
- Add new product listings.
- View detailed product information.
- Edit or delete own products.
- User profile management.
- MongoDB-based data storage.
- RESTful API communication.

7.3 Advantages of the Proposed System

The proposed system offers several advantages over traditional methods:

- Dedicated platform for college students.
- Secure authentication using JWT.
- Organized product management.
- Easy navigation and user-friendly interface.
- Faster product discovery.
- Reliable database management.

- Centralized marketplace.
- Scalable architecture for future enhancements.

7.4 Existing System vs Proposed System

Feature	Existing System	ReNest (Proposed System)
Platform	WhatsApp, Facebook, OLX	Dedicated Student Marketplace
User Authentication	Not Available	JWT Authentication
User Profile	Limited	Available
Product Management	Manual	Complete CRUD Operations
Product Categories	Limited	Organized Categories
Database	Not Available	MongoDB
Security	Low	High
Search & Browsing	Difficult	Easy
Product Updates	Manual	Instant
Scalability	Limited	High

7.5 Benefits of ReNest

The implementation of ReNest provides the following benefits:

For Students

- Easy buying and selling process.
- Trusted campus-based marketplace.
- Affordable access to second-hand products.
- Better product visibility.
- Reduced dependency on social media.

For Sellers

- Easy product management.
- Edit and delete listings anytime.
- Better reach within the college community.
- Organized product information.

For the Institution

- Encourages sustainable reuse of resources.
- Promotes digital learning initiatives.
- Supports a safer and more organized exchange platform.

7.6 Future Enhancements

The current version of ReNest provides the core marketplace functionalities. The following features can be added in future versions:

- Real-time chat between buyers and sellers.
- Wishlist management.
- Product search with advanced filters.
- Image upload using cloud storage.
- Ratings and reviews.
- College email verification.
- Online payment gateway.
- Push notifications.
- Admin dashboard.
- Mobile application support.

7.7 Outcome of the Proposed System

The proposed system successfully addresses the limitations of existing buying and selling methods by providing a centralized, secure, and user-friendly marketplace exclusively for students. It simplifies product management, improves accessibility, and creates a trusted environment for campus transactions.

Chapter Summary

This chapter presented the proposed solution, ReNest, and explained how it improves upon existing methods of buying and selling second-hand products. It also highlighted the key advantages, comparison with the existing system, benefits, and future enhancements.

CHAPTER 8
REQUIREMENT ANALYSIS

8.1 Introduction

Requirement analysis is the process of identifying the functional and non-functional requirements of a software system before its development. It helps in understanding the system objectives, user expectations, hardware and software needs, and overall project feasibility.

For the development of ReNest, the requirements were analyzed to ensure that the application provides a secure, efficient, and user-friendly marketplace for students.

8.2 Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or above
RAM	4 GB
Storage	10 GB Free Disk Space
Display	1366 × 768 Resolution
Keyboard	Standard Keyboard
Mouse	Standard Mouse
Internet	Required

8.3 Software Requirements

Software	Purpose
Windows 11	Operating System
Visual Studio Code	Code Editor
Node.js	Backend Runtime
Express.js	Backend Framework
MongoDB	Database

Software	Purpose
Postman	API Testing
Git	Version Control
GitHub	Source Code Repository
Google Chrome	Application Testing

8.4 Functional Requirements

The functional requirements define the features provided by the system.

User Module

- User Registration
- User Login
- User Logout
- User Profile Management

Product Module

- Add Product
- View Product
- Edit Product
- Delete Product
- Browse Marketplace

Authentication Module

- JWT Authentication
- Password Hashing
- Protected Routes

Database Module

- Store User Data
- Store Product Data
- CRUD Operations

API Module

- Authentication APIs
- Product APIs

- User APIs
- Middleware Integration

8.5 Non-Functional Requirements

Requirement	Description
Performance	Fast response time for user requests
Security	JWT Authentication and encrypted passwords
Reliability	Stable API responses and secure data handling
Availability	Accessible through a web browser
Maintainability	Modular code structure for easy updates
Scalability	Can support future feature expansion
Usability	Simple and intuitive user interface

8.6 User Requirements

The application should allow users to:

- Create an account.
- Log in securely.
- Browse available products.
- View product details.
- Add new product listings.
- Edit or delete their own products.
- Manage their profile.
- Contact sellers.

8.7 System Requirements

The system should:

- Authenticate users securely.
- Validate user inputs.
- Store data in MongoDB.

- **Handle API requests efficiently.**
- **Prevent unauthorized access.**
- **Provide proper error handling.**
- **Maintain data consistency.**
- **Support future enhancements.**

8.8 Constraints

The current implementation has the following constraints:

- **Internet connection is required.**
- **Product image upload is limited in the current version.**
- **No integrated payment gateway.**
- **Communication between buyer and seller is handled outside the application.**
- **The platform is intended for student use within a campus environment.**

CHAPTER 9

SYSTEM DESIGN

9.1 Introduction

System design is the process of defining the architecture, components, modules, database structure, and interactions of a software system. It provides a blueprint for implementation and ensures that the application is scalable, maintainable, and easy to understand.

For ReNest, a layered architecture was adopted to separate the presentation layer, business logic, and database operations.

9.2 System Architecture

The ReNest application follows a three-tier architecture, where each layer has a specific responsibility.

Layers of the System

1. Presentation Layer

This layer represents the user interface of the application. It is developed using:

- HTML5
- CSS3
- JavaScript

Users interact with the application through web pages such as Login, Marketplace, Product Details, Dashboard, and Profile.

---

2. Application Layer

The application layer contains the business logic of the system.

It is developed using:

- Node.js
- Express.js

This layer is responsible for:

- Processing client requests
- User authentication
- Product management

- API handling
- Input validation
- Middleware execution
- Error handling

---

3. Database Layer

The database layer stores all application data.

It uses:

- MongoDB
- Mongoose

This layer manages:

- User information
- Product listings
- Authentication data
- Database queries

Figure 9.1 System Architecture

```
+----------------------+
|       User          |
+----------+-----------+
           |
           |
  HTTP Request/Response
           |
           ▼
+---------------------------------------+
|    Frontend (Presentation Layer)     |
|---------------------------------------|
| HTML | CSS | JavaScript              |
+---------------+-----------------------+
                |
                | REST API
                ▼
+---------------------------------------+
| Backend (Application Layer)          |
|---------------------------------------|
| Node.js                              |
| Express.js                           |
| JWT Authentication                   |
| Middleware                           |
| Controllers                          |
+---------------+-----------------------+
                |
                |
                ▼
+---------------------------------------+
| Database Layer                       |
|---------------------------------------|
| MongoDB                              |
| Mongoose                             |
+---------------------------------------+
```

Explanation of the Architecture

The user accesses the application through a web browser and interacts with the frontend developed using HTML, CSS, and JavaScript. Whenever the user performs an action, such as logging in or adding a product, the frontend sends an HTTP request to the backend server.

The backend, developed using Node.js and Express.js, processes the request, performs authentication and validation, and communicates with the MongoDB database using Mongoose. After processing the request, the backend sends an appropriate response back to the frontend, which is then displayed to the user.

This layered architecture ensures better modularity, easier maintenance, and improved scalability.

Advantages of the Architecture

- Separation of concerns.
- Modular development.
- Easy debugging and maintenance.
- Better security through middleware.
- Scalable backend architecture.
- Efficient database management.
- Simplified API communication.

9.3 System Workflow

The workflow of ReNest illustrates how a user interacts with the application from authentication to product management. The process ensures that only authenticated users can access protected features while maintaining secure communication between the frontend, backend, and database.

Workflow Steps

1. User opens the ReNest application.
2. User registers or logs into the system.
3. The server validates the user's credentials.
4. Upon successful authentication, a JWT token is generated.
5. The authenticated user can browse available products.
6. The user can add, edit, or delete their own product listings.
7. Product data is stored and retrieved from the MongoDB database.

8. The requested information is displayed to the user.
9. The user logs out, ending the session.

Figure 9.2 Workflow of ReNest

Start

|

▼

Open ReNest Website

|

▼

Register / Login Account

|

▼

User Credentials Valid?

No | Yes

Display Error | Generate JWT

|

▼

Browse Marketplace

|

▼

View Product Information

|

▼

Add / Edit / Delete Product

|

▼

Update MongoDB Database

|

▼

Display Updated Data

```
|
▼
Logout
|
▼
End
```

Figure 9.2: Workflow of ReNest

Workflow Explanation

The application begins when a user visits the ReNest website. If the user does not have an account, they must complete the registration process. Existing users can log in using their registered credentials.

After successful authentication, the backend generates a JWT token that is used to authorize future requests. Authenticated users can browse products, view product details, and manage their own listings. All product-related operations are processed by the backend and stored in the MongoDB database. Finally, the user can log out, ending the authenticated session.

9.4 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) represents how information moves between users, the application, and the database.

Level 0 DFD (Context Diagram)

```
+----------------+
|   Student   |
+--------+-------+
|
Login / Product Requests
|
▼
+------------------+
|    ReNest    |
|  Application  |
+--------+---------+
```

```
           |
Product & User Information
           |
           ▼
  +------------------+
  |    MongoDB     |
  +------------------+
```

Figure 9.3: Level 0 Data Flow Diagram

Explanation

The student interacts with the ReNest application by sending requests such as registration, login, product browsing, and product management. The application processes these requests and stores or retrieves the required information from the MongoDB database before returning the response to the user.

Level 1 DFD

```
            Student
               |
               ▼
+-------------------------+
| Authentication Module   |
+------------+------------+
             |
             ▼
       User Collection
             |
             ▼
+-------------------------+
| Product Management      |
+------------+------------+
             |
             ▼
     Product Collection
```

```
            |
            ▼
+--------------------------+
| API Response             |
+------------+-------------+
             |
             ▼
          Student
```

Figure 9.4: Level 1 Data Flow Diagram

Explanation

The Level 1 DFD illustrates the internal functioning of the application. User authentication is performed first, after which authenticated users can access product management functionalities. Product information is stored in the Product Collection, while user details remain in the User Collection. The backend processes requests and returns appropriate responses to the frontend.

Advantages of the DFD

- Clearly represents data movement.
- Simplifies system understanding.
- Helps identify data sources and destinations.
- Improves communication during system design.
- Useful for future maintenance and enhancements.

9.5 Flowchart

The flowchart illustrates the logical flow of the ReNest application from user authentication to product management.

Figure 9.5 Flowchart of ReNest

```
START
  |
  ▼
Open ReNest Website
  |
  ▼
Register / Login
  |
  ▼
Credentials Verified?
```

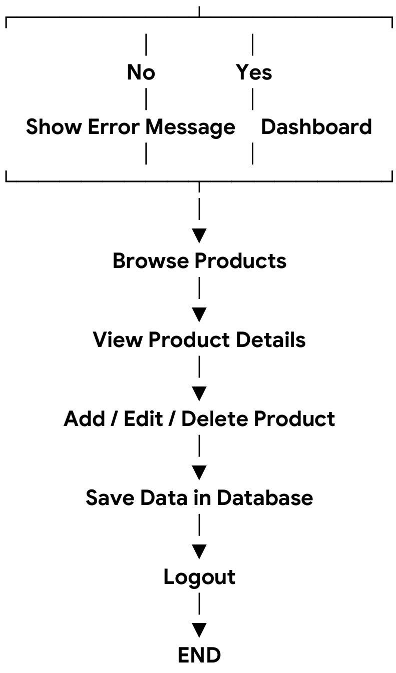


Figure 9.5: **Flowchart of ReNest**

Explanation

The flowchart begins with the user accessing the website. After successful authentication, the user is redirected to the dashboard, where marketplace operations such as browsing products and managing listings can be performed. All changes are stored in the database before the user logs out.

9.6 Use Case Diagram

A Use Case Diagram represents the interaction between the user and the ReNest application.

Primary Actor

Student (User)

Use Cases

- **Register**
- **Login**
- **Browse Marketplace**
- **View Product Details**
- **Add Product**
- **Edit Product**
- **Delete Product**
- **Manage Profile**
- **Logout**

Figure 9.6 Use Case Diagram

```
        +----------------+
        |   Student    |
        +-------+--------+
                |
  ---------------------------------------------
     |   |   |   |   |   |   |
Register Login Browse View Add Edit Delete Profile Logout
```

Figure 9.6: **Use Case Diagram**

Explanation

The Student is the primary actor of the system. After logging in, the student can browse products, view product details, manage personal listings, update profile information, and securely log out of the application.

9.7 Class Diagram

The Class Diagram provides a conceptual representation of the major entities used in the application.

Figure 9.7 Class Diagram

```
+----------------------+
|      User       |
+----------------------+
| _id            |
| name           |
| email          |
| password       |
| phone          |
+----------------------+

          |
        | owns
          ▼

+----------------------+
|    Product      |
+----------------------+
| _id            |
| title          |
| category       |
| description    |
| price          |
| image          |
| ownerId        |
+----------------------+
```

Figure 9.7: **Class Diagram**

Explanation

The User **class stores authentication and profile information, while the** Product **class stores all marketplace listings. A single user can own multiple products, establishing a one-to-many relationship.**

9.8 Activity Diagram

The Activity Diagram illustrates the sequence of activities performed during user interaction.

Figure 9.8 Activity Diagram

```
        Start
          |
          ▼
     Open Website
          |
          ▼
    Login/Register
          |
          ▼
    Authentication
          |
          ▼
      Dashboard
          |
          ▼
   Browse Products
          |
          ▼
   Manage Products
          |
          ▼
        Logout
          |
          ▼
         End
```

Figure 9.8: **Activity Diagram**

Explanation

The activity begins when a user accesses the website. After authentication, the user can browse the marketplace, perform product management operations, and finally log out, ending the session.

9.9 Sequence Diagram

The Sequence Diagram shows the communication between different components of the system.

Figure 9.9 Sequence Diagram

```
User
 |
 |  Login Request
 ▼
Frontend
 |
 |  API Request
 ▼
Backend
 |
 |  Verify User
 ▼
MongoDB
 |
 |  User Data
 ▲
Backend
 |
 |  JWT Token
 ▲
Frontend
 |
 |  Dashboard
 ▲
User
```

Figure 9.9: **Sequence Diagram**

Explanation

The user submits login credentials through the frontend. The backend verifies these credentials using MongoDB. If authentication is successful, a JWT token is generated and returned to the frontend, allowing the user to access protected features.

9.10 Entity Relationship (ER) Diagram

The ER Diagram represents the relationship between the main database entities.

Figure 9.10 ER Diagram

```
+-----------+      1     M     +------------+
|  User    |-----------------------|  Product  |
+-----------+                  +------------+
| userId    |                  | productId  |
| name      |                  | title      |
| email     |                  | category   |
| password  |                  | price      |
| phone     |                  | ownerId FK |
+-----------+                  +------------+
```

Figure 9.10: **Entity Relationship Diagram**

Explanation

Each User **can create multiple** Product **listings. The ownerId field in the Product collection acts as a foreign reference to the User collection, establishing a** one-to-many relationship**.**

Chapter Summary

This chapter presented the complete system design of ReNest**, including the architecture, workflow, flowchart, DFD, Use Case Diagram, Class Diagram, Activity Diagram, Sequence Diagram, and Entity Relationship Diagram. These diagrams collectively describe the structure, functionality, and interactions within the application, providing a clear blueprint for implementation.**

CHAPTER 10

IMPLEMENTATION

10.1 Introduction

Implementation is the phase in which the system design is converted into a working software application. During this phase, the frontend interface, backend services, database connectivity, authentication mechanisms, and REST APIs were developed and integrated to build the complete ReNest application.

The project follows a modular approach, where different functionalities are divided into separate components, making the application easier to understand, maintain, and enhance.

10.2 Project Structure

The ReNest project is organized into different folders based on their functionality. This modular structure improves code readability and simplifies future maintenance.

Project Structure

```
ReNest/
│
├── client/
│   ├── assets/
│   ├── css/
│   ├── images/
│   ├── js/
│   └── pages/
│
├── server/
│   ├── config/
│   ├── controllers/
│   ├── middleware/
│   ├── models/
│   ├── routes/
│   ├── utils/
│   └── server.js
│
├── .env
├── package.json
└── README.md
```

Figure 10.1: Project Folder Structure

(Modify the folder names if your final project structure differs.)

10.3 Folder Description

Folder/File	Purpose
client	Contains all frontend files
assets	Stores images, icons, and static resources
css	Stylesheets for different pages
js	JavaScript files for frontend functionality
pages	HTML pages of the application
server	Contains backend source code
config	Database configuration
controllers	Business logic of APIs
middleware	Authentication and validation middleware
models	MongoDB schemas and models
routes	REST API endpoints
utils	Helper functions
.env	Stores environment variables
package.json	Project dependencies and scripts

10.4 Frontend Implementation

The frontend of ReNest was developed using HTML5, CSS3, and JavaScript. It provides an interactive user interface through which users can access all features of the application.

Major Frontend Pages

Page	Purpose
Home Page	Landing page of the application
Login Page	User authentication

Page	Purpose
Registration Page	New user registration
Marketplace	Display available products
Product Details	Shows complete information about a product
Sell Product	Add a new product
Dashboard	Manage user activities
Profile	Display and update user information

10.5 Backend Implementation

The backend was developed using Node.js and Express.js. It manages application logic, authentication, database communication, and REST API handling.

The backend performs the following operations:

- User authentication.
- Product management.
- Request validation.
- Database operations.
- Error handling.
- Protected route management.

10.6 Database Implementation

MongoDB was used as the primary database for storing application data.

The database stores:

- User Information
- Product Information
- Authentication Data

Mongoose was used to define schemas and perform CRUD operations between the application and the MongoDB database.

10.7 Development Workflow

The implementation followed the following workflow:

1. Requirement Analysis
2. Frontend Design
3. Backend Development
4. Database Integration
5. API Development
6. Authentication Implementation
7. Testing
8. Bug Fixing
9. Documentation

10.8 User Authentication

User authentication is one of the core functionalities of ReNest. It ensures that only registered users can access protected features such as adding products, editing listings, and managing personal information.

The authentication process consists of the following steps:

1. The user registers by providing the required information.
2. The password is encrypted using bcrypt before being stored.
3. During login, the entered password is compared with the encrypted password.
4. If the credentials are valid, the server generates a JSON Web Token (JWT).
5. The generated token is returned to the client.
6. The token is attached to protected requests for user verification.

10.9 Authentication Flow

```
User Login
    |
    ▼
Validate Credentials
    |
    ▼
Compare Password (bcrypt)
    |
    ▼
Generate JWT Token
```

|

▼

Return Token

|

▼

Access Protected Routes

Figure 10.2: Authentication Flow

10.10 REST API Design

The backend communicates with the frontend using RESTful APIs. Each API performs a specific operation and returns responses in JSON format.

Authentication APIs

Method	Endpoint	Purpose
POST	/register	Register a new user
POST	/login	Authenticate user
GET	/profile	Fetch user profile

Product APIs

Method	Endpoint	Purpose
GET	/products	Retrieve all products
GET	/products/:id	View product details
POST	/products	Add a new product
PUT	/products/:id	Update product
DELETE	/products/:id	Delete product

Note: Replace the endpoints with your actual API routes if they differ.

10.11 Database Design

The application stores its data in MongoDB using two primary collections.

User Collection

Field	Description
Name	User's full name
Email	User email address
Password	Encrypted password
Phone	Contact number

Product Collection

Field	Description
Title	Product title
Category	Product category
Description	Product description
Price	Selling price
Image	Product image
Owner ID	Reference to the user

10.12 Middleware

Middleware acts as an intermediary between the client request and the server response.

The middleware used in ReNest performs the following functions:

- JWT verification.
- Request authentication.
- Input validation.
- Error handling.
- Route protection.

Using middleware improves security and prevents unauthorized users from accessing restricted resources.

10.13 Controllers

Controllers contain the business logic of the application. They process incoming requests, interact with the database, and send appropriate responses to the client.

The major controller responsibilities include:

- User registration.
- User login.
- Product management.
- Profile management.
- Response generation.

Separating controllers from routes improves project organization and maintainability.

10.14 Routes

Routes define the API endpoints of the application and determine how client requests are handled.

The routes are categorized into:

- Authentication Routes
- Product Routes
- User Routes

Each route forwards the request to its respective controller for processing.

10.15 Models

Models define the structure of the MongoDB collections using Mongoose schemas.

The project primarily consists of:

- User Model
- Product Model

These models ensure consistent storage and validation of data within the database.

10.16 Error Handling

The application includes error handling mechanisms to improve stability and user experience.

Examples include:

- Invalid login credentials.
- Missing required fields.
- Unauthorized access.
- Invalid product details.
- Database connection errors.
- Internal server errors.

Proper error messages help users understand and resolve issues efficiently.

10.17 Security Features

To enhance application security, the following measures have been implemented:

- Password hashing using bcrypt.
- JWT-based authentication.
- Protected API routes.
- Input validation.
- Environment variable management using dotenv.
- Secure database communication.

10.18 Implementation Summary

The implementation of ReNest combines frontend technologies, backend services, and database management into a single integrated system. The modular architecture, secure authentication, RESTful APIs, and MongoDB database ensure that the application is maintainable, scalable, and suitable for future enhancements.

Chapter Summary

This chapter explained the implementation details of ReNest, including project organization, authentication, REST APIs, database design, middleware, controllers, models, routes, and security features. The modular implementation simplifies development while ensuring secure and efficient communication between different components of the application.

CHAPTER 11

TESTING

11.1 Introduction

Testing is an essential phase of software development that ensures the application functions correctly and meets the specified requirements. During the development of ReNest, different testing techniques were performed to verify the functionality, security, and reliability of the application.

The objective of testing was to identify errors, validate user inputs, verify API responses, and ensure smooth interaction between the frontend, backend, and database.

11.2 Objectives of Testing

The main objectives of testing were:

- To verify that all modules work correctly.
- To detect and fix software bugs.
- To validate user inputs.
- To ensure secure authentication.
- To verify API functionality.
- To improve application reliability.
- To ensure proper database operations.

11.3 Types of Testing Performed

Testing Type	Purpose
Unit Testing	Testing individual functions and modules
Integration Testing	Verifying communication between frontend, backend, and database
Functional Testing	Checking whether application features work correctly
API Testing	Testing REST APIs using Postman
Validation Testing	Verifying user input and error handling

11.4 Test Cases

Test Case	Input	Expected Result	Status
User Registration	Valid user details	Account created successfully	Pass
User Registration	Existing email	Error message displayed	Pass
User Login	Correct credentials	Login successful	Pass
User Login	Invalid credentials	Login failed	Pass
Add Product	Valid product details	Product added successfully	Pass
Edit Product	Updated information	Product updated	Pass
Delete Product	Existing product	Product removed	Pass
Browse Products	Request all products	Product list displayed	Pass
Protected Route	Invalid JWT Token	Access denied	Pass
User Profile	Valid user	Profile displayed	Pass

11.5 API Testing

The REST APIs were tested using Postman.

The following APIs were verified:

- User Registration API
- User Login API
- Product Listing API
- Product Details API
- Update Product API
- Delete Product API
- User Profile API

[Insert Screenshot: Postman API Testing]

11.6 Input Validation

The application validates user input before processing requests.

The following validations were implemented:

- Required fields cannot be empty.
- Email format validation.
- Password validation.
- Product title validation.
- Product price validation.
- Unauthorized access prevention using JWT.
- Invalid request handling.

11.7 Expected Results

The expected behavior of the application includes:

- Successful user authentication.
- Proper product management.
- Accurate database operations.
- Secure API communication.
- Correct error handling.
- Stable application performance.

11.8 Actual Results

After testing, the following observations were made:

- All authentication features worked successfully.
- Product CRUD operations were executed correctly.
- MongoDB stored and retrieved data accurately.
- Protected routes were accessible only to authenticated users.
- API responses were generated correctly.
- Error messages were displayed appropriately for invalid requests.

11.9 Advantages of Testing

- **Improved software quality.**
- **Reduced application errors.**
- **Increased system reliability.**
- **Better user experience.**
- **Enhanced application security.**
- **Easier debugging and maintenance.**

Chapter Summary

This chapter discussed the testing process carried out during the development of ReNest. Various testing techniques, including unit testing, integration testing, functional testing, and API testing, were performed to verify the correctness, security, and reliability of the application. The results confirmed that the implemented features functioned as expected and met the project requirements.

CHAPTER 12

INTERNSHIP EVALUATION SHEET

12.1 Student Details

Particular	Details
Student Name	Utkarsh Yadav
Roll Number	2403611530068
Course	Bachelor of Technology
Branch	Computer Science & Engineering (Artificial Intelligence & Machine Learning)
College	R.R. Institute of Modern Technology, Lucknow
University	Dr. A.P.J. Abdul Kalam Technical University (AKTU)
Internship Duration	40 Days
Project Title	ReNest – A Full Stack Student Marketplace Web Application
Faculty Guide	Mr. Bhavesh Mathur

12.2 Performance Evaluation

Evaluation Criteria	Maximum Marks	Marks Obtained
Attendance & Punctuality	10	
Technical Knowledge	10	
Programming Skills	15	
Project Planning	10	
Project Implementation	15	
Documentation	10	
Problem Solving Ability	10	
Communication Skills	10	
Presentation	10	
Overall Performance	10	

Total	**100**	

12.3 Faculty Remarks

Remarks:

12.4 Recommendation

 Excellent

 Very Good

 Good

 Satisfactory

 Needs Improvement

12.5 Signatures

Faculty Guide	**Head of Department**
Signature: ____________	**Signature: ____________**
Mr. Bhavesh Mathur	**HOD, CSE (AI&ML) Department**
Date: ____________	**Date: ____________**

Chapter Summary

This chapter provides the internship evaluation sheet used to assess the student's overall performance during the internship. The evaluation is based on technical knowledge, project implementation, documentation, communication skills, and overall professional development.

CHAPTER 13

DAILY WORK REPORT

13.1 Introduction

The Summer Internship was conducted over a period of 40 days, during which theoretical concepts and practical implementation were carried out simultaneously. The initial phase focused on learning Full Stack Web Development technologies, while the later phase involved designing, developing, testing, and documenting the ReNest project.

13.2 40-Day Internship Work Report

Day	*Work Performed*
Day 1	Internship orientation and introduction to Full Stack Web Development.
Day 2	Learned HTML structure, semantic elements, and webpage creation.
Day 3	Practiced HTML concepts and built basic web pages.
Day 4	Studied HTML forms, tables, lists, and multimedia elements.
Day 5	Learned CSS fundamentals and page styling techniques.
Day 6	Implemented responsive layouts using Flexbox and Grid.
Day 7	Explored JavaScript basics, variables, operators, and functions.
Day 8	Learned DOM manipulation and event handling.
Day 9	Studied asynchronous JavaScript concepts and API handling.
Day 10	Introduction to Node.js and backend development.
Day 11	Learned Express.js routing and middleware concepts.
Day 12	Studied MongoDB and basic CRUD operations.
Day 13	Implemented MongoDB connection using Mongoose.
Day 14	Learned Git, GitHub, and version control workflow.
Day 15	Requirement analysis and planning of the ReNest project.
Day 16	Designed the project architecture and folder structure.
Day 17	Created frontend pages for the application.
Day 18	Designed the user interface using CSS.
Day 19	Implemented user registration functionality.

Day 20	Developed secure user login using JWT authentication.
Day 21	Integrated bcrypt for password encryption.
Day 22	Created MongoDB models for users and products.
Day 23	Developed REST APIs for authentication.
Day 24	Implemented product listing functionality.
Day 25	Developed product details page and data retrieval APIs.
Day 26	Implemented edit and delete product features.
Day 27	Developed user profile management module.
Day 28	Integrated frontend with backend APIs.
Day 29	Tested API endpoints using Postman.
Day 30	Implemented middleware for route protection.
Day 31	Added input validation and improved error handling.
Day 32	Performed debugging and resolved application issues.
Day 33	Improved UI design and user experience.
Day 34	Optimized backend API performance.
Day 35	Conducted integration testing of all modules.
Day 36	Fixed identified bugs and improved application stability.
Day 37	Verified authentication and database operations.
Day 38	Prepared project documentation and diagrams.
Day 39	Performed final testing and project review.
Day 40	Completed internship report and final project submission.

13.3 Overall Outcome

During the internship, practical exposure was gained in modern web development technologies and software engineering practices. The development of ReNest helped strengthen knowledge of frontend development, backend development, database management, RESTful APIs, authentication, testing, debugging, and project documentation.

CHAPTER 14
CHALLENGES FACED

14.1 Introduction

During the development of ReNest, several technical and implementation challenges were encountered. These challenges provided valuable learning opportunities and helped improve problem-solving and debugging skills. Each issue was analyzed systematically and resolved using appropriate development practices.

14.2 Challenges and Solutions

CHALLENGE	SOLUTION
UNDERSTANDING THE EXPRESS.JS FRAMEWORK AND ROUTING	**Studied Express documentation and implemented routes through practical examples.**
ESTABLISHING A CONNECTION WITH MONGODB	**Configured MongoDB properly using Mongoose and verified the database connection.**
IMPLEMENTING JWT AUTHENTICATION	**Learned the JWT authentication flow and protected routes using authentication middleware.**
PASSWORD SECURITY	**Used bcrypt to hash passwords before storing them in the database.**
DESIGNING THE DATABASE SCHEMA	**Planned separate collections for users and products to maintain proper relationships.**
API DEVELOPMENT	**Designed RESTful APIs and tested them using Postman before frontend integration.**
FRONTEND AND BACKEND INTEGRATION	**Connected frontend pages with backend APIs using JavaScript and verified API responses.**
INPUT VALIDATION	**Added validation checks for user registration, login, and product details to prevent invalid data.**
ERROR HANDLING	**Implemented proper error messages and handled exceptions to improve user experience.**
PROJECT ORGANIZATION	**Divided the project into modules such as routes, models, controllers, and middleware for better maintainability.**

14.3 Technical Challenges

Some of the major technical challenges included:

- **Understanding REST API communication.**
- **Managing JWT-based authentication.**
- **Working with MongoDB collections.**
- **Organizing backend files efficiently.**
- **Debugging asynchronous JavaScript code.**
- **Testing APIs before frontend integration.**

14.4 Lessons Learned

These challenges helped in:

- **Improving debugging skills.**
- **Understanding backend architecture.**
- **Learning secure authentication techniques.**
- **Gaining practical experience in API development.**
- **Developing confidence in building full-stack applications.**

Chapter Summary

This chapter discussed the major challenges encountered during the development of ReNest and the solutions adopted to overcome them. The experience gained from resolving these issues contributed significantly to improving technical knowledge and practical software development skills.

CHAPTER 15

SKILLS GAINED

15.1 Introduction

The internship provided practical exposure to modern software development technologies and strengthened both technical and professional competencies. The knowledge gained during the training was applied throughout the development of the ReNest project.

15.2 Technical Skills

The following technical skills were developed during the internship:

- HTML5
- CSS3
- JavaScript
- Node.js
- Express.js
- MongoDB
- Mongoose
- REST API Development
- JWT Authentication
- Password Hashing using bcrypt
- Git and GitHub
- API Testing using Postman
- Database Design
- Debugging and Troubleshooting
- Software Documentation

15.3 Professional Skills

The internship also helped improve the following professional skills:

- Problem Solving
- Logical Thinking

- Time Management
- Team Collaboration
- Communication Skills
- Technical Documentation
- Analytical Thinking
- Software Development Planning
- Project Management

15.4 Internship Outcome

By the end of the internship, I was able to understand the complete workflow of a Full Stack Web Application, including frontend development, backend development, database management, authentication, testing, and documentation. The internship also enhanced my confidence in developing real-world software projects using modern technologies.

Chapter Summary

This chapter highlighted the technical and professional skills acquired during the internship. These skills will be valuable for future academic projects, internships, and professional software development roles.

CHAPTER 16

CONCLUSION

16.1 Conclusion

The Summer Internship provided an excellent opportunity to gain practical experience in Full Stack Web Development and understand the complete software development process. Throughout the internship, modern technologies such as HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, JWT Authentication, and RESTful APIs were learned and applied in the development of the ReNest project.

The project successfully achieved its objective of providing a secure and user-friendly marketplace for college students to buy and sell second-hand products within their campus community. The modular architecture, secure authentication mechanism, organized database design, and RESTful API implementation make the application reliable, scalable, and suitable for future enhancements.

In addition to technical knowledge, the internship improved problem-solving abilities, debugging skills, project planning, teamwork, and technical documentation. It also provided valuable insight into real-world software development practices and strengthened my confidence in developing full-stack web applications.

Overall, the internship was a valuable learning experience that helped bridge the gap between academic knowledge and practical implementation while contributing significantly to my professional and technical growth.

REFERENCES

Books

1. Ethan Brown, *Web Development with Node and Express*, O'Reilly Media.
2. Brad Dayley, Brendan Dayley, Caleb Dayley, *Node.js, MongoDB and Angular Web Development*, Addison-Wesley.
3. Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, *MongoDB: The Definitive Guide*, O'Reilly Media.
4. Jon Duckett, *HTML and CSS: Design and Build Websites*, John Wiley & Sons.
5. Jon Duckett, *JavaScript and jQuery: Interactive Front-End Web Development*, John Wiley & Sons.
6. Ian Sommerville, *Software Engineering*, 10th Edition, Pearson Education.

Official Documentation

1. HTML Living Standard – https://html.spec.whatwg.org/
2. MDN Web Docs (HTML, CSS & JavaScript) – https://developer.mozilla.org/
3. Node.js Documentation – https://nodejs.org/docs/latest/api/
4. Express.js Documentation – https://expressjs.com/
5. MongoDB Documentation – https://www.mongodb.com/docs/
6. Mongoose Documentation – https://mongoosejs.com/docs/
7. JSON Web Token (JWT) Documentation – https://jwt.io/
8. bcrypt Documentation – https://github.com/kelektiv/node.bcrypt.js
9. Dotenv Documentation – https://github.com/motdotla/dotenv
10. Postman Learning Center – https://learning.postman.com/

Development Tools

- Visual Studio Code
- Node.js
- MongoDB Community Server / MongoDB Atlas
- Git
- GitHub
- Postman

- Google Chrome

Online Resources

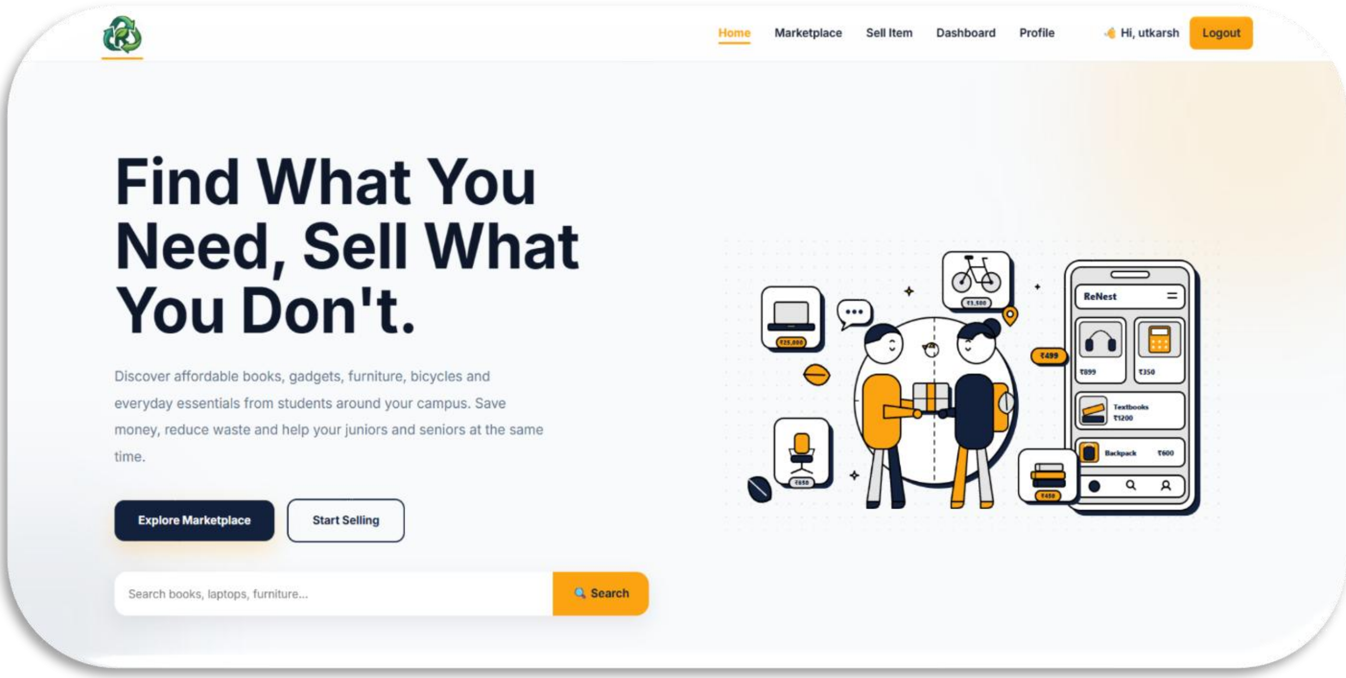
- W3Schools – https://www.w3schools.com/
- GeeksforGeeks – https://www.geeksforgeeks.org/
- Stack Overflow – https://stackoverflow.com/

APPENDIX

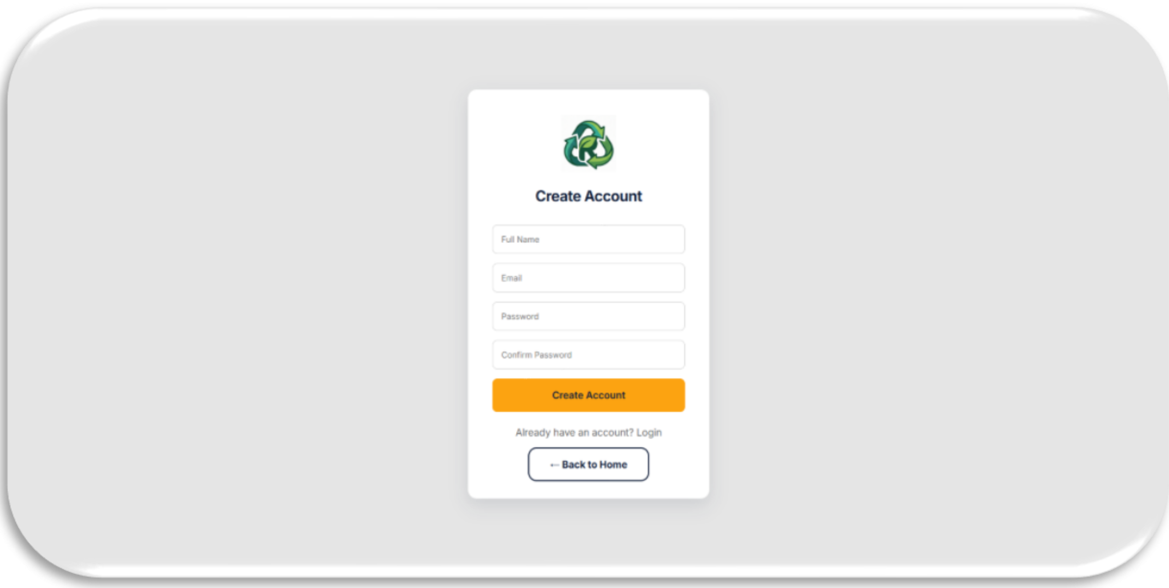
The appendix contains supporting material related to the development and testing of the ReNest project. These materials provide additional evidence of the implementation process and are included for reference.

Appendix A: Project Screenshots

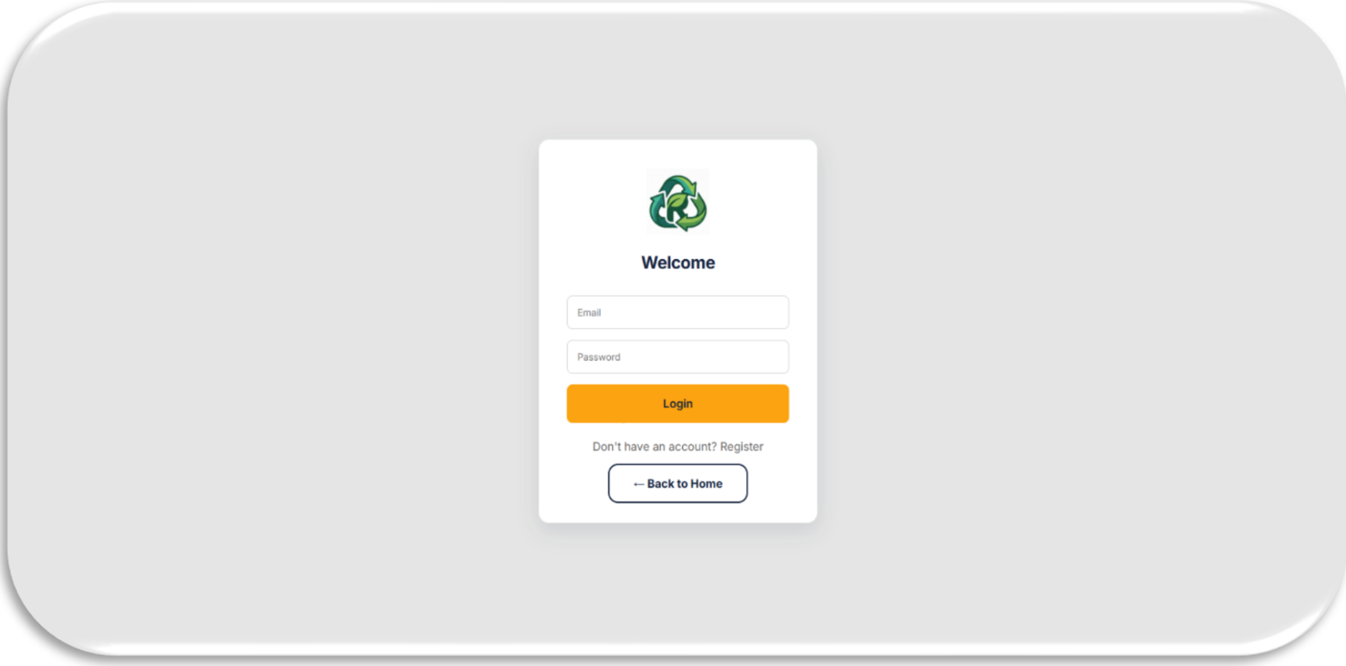
A.1 Home Page



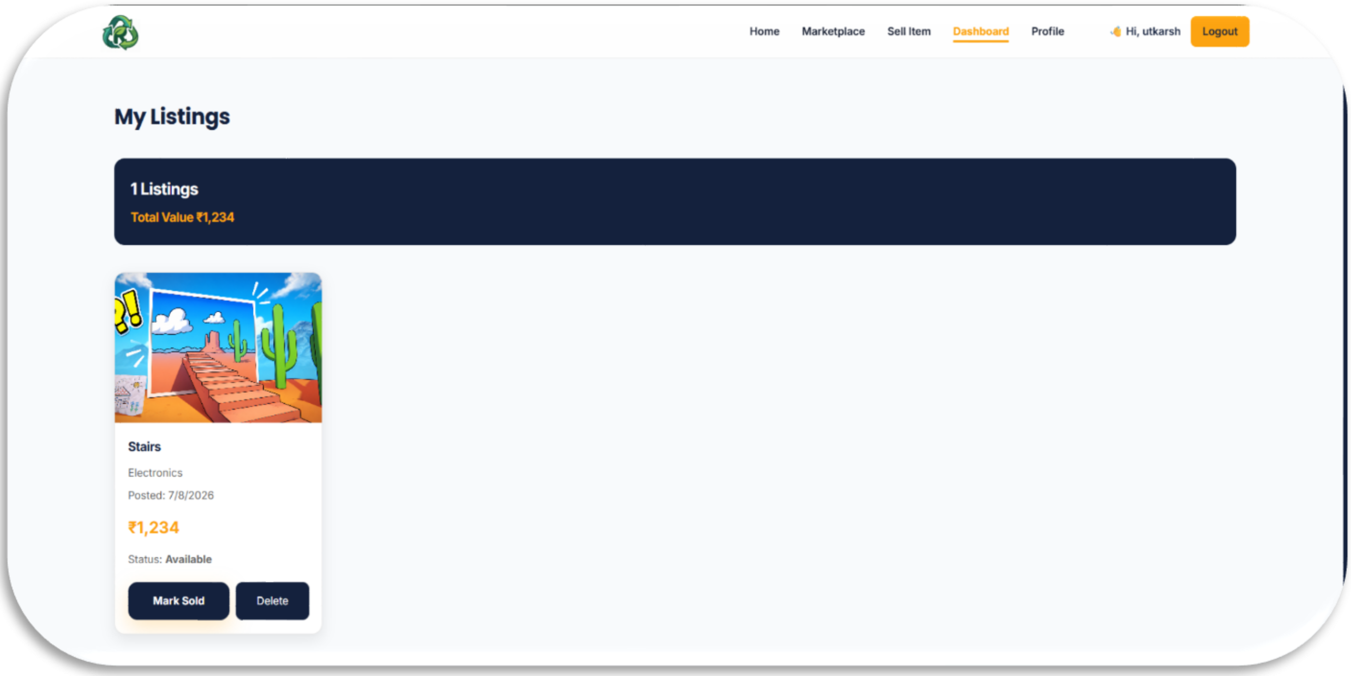
A.2 User Registration



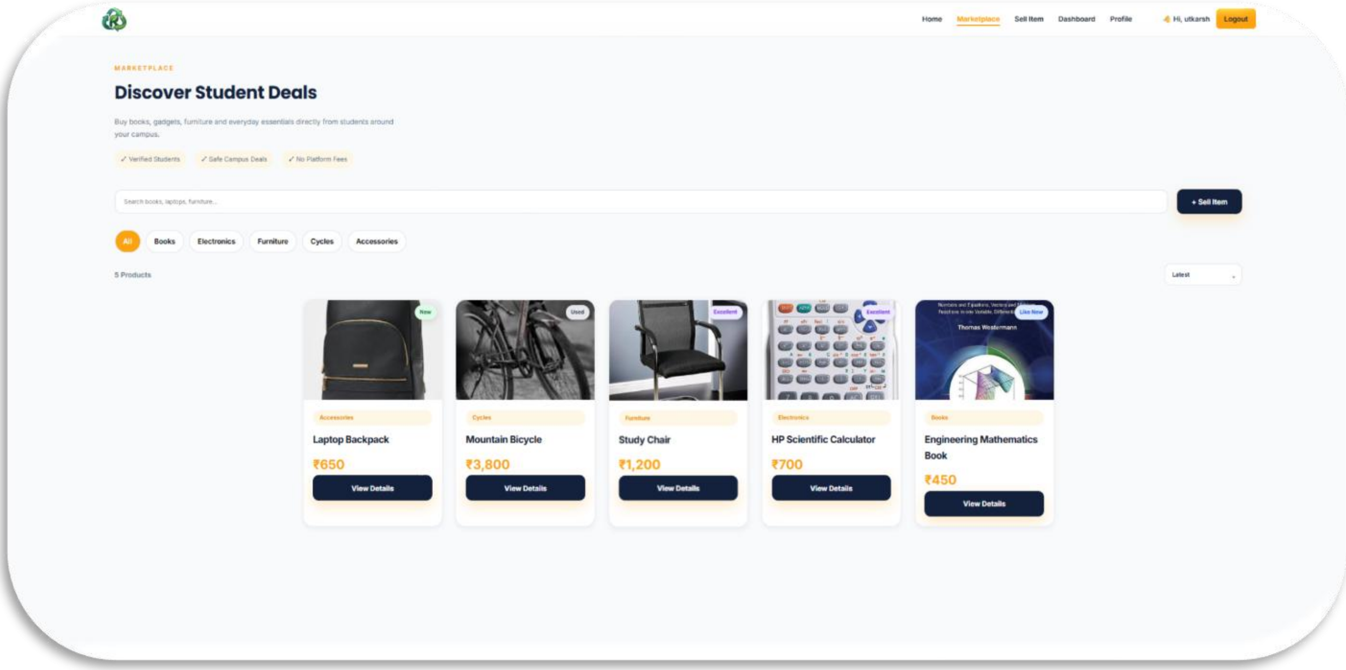
A.3 User Login



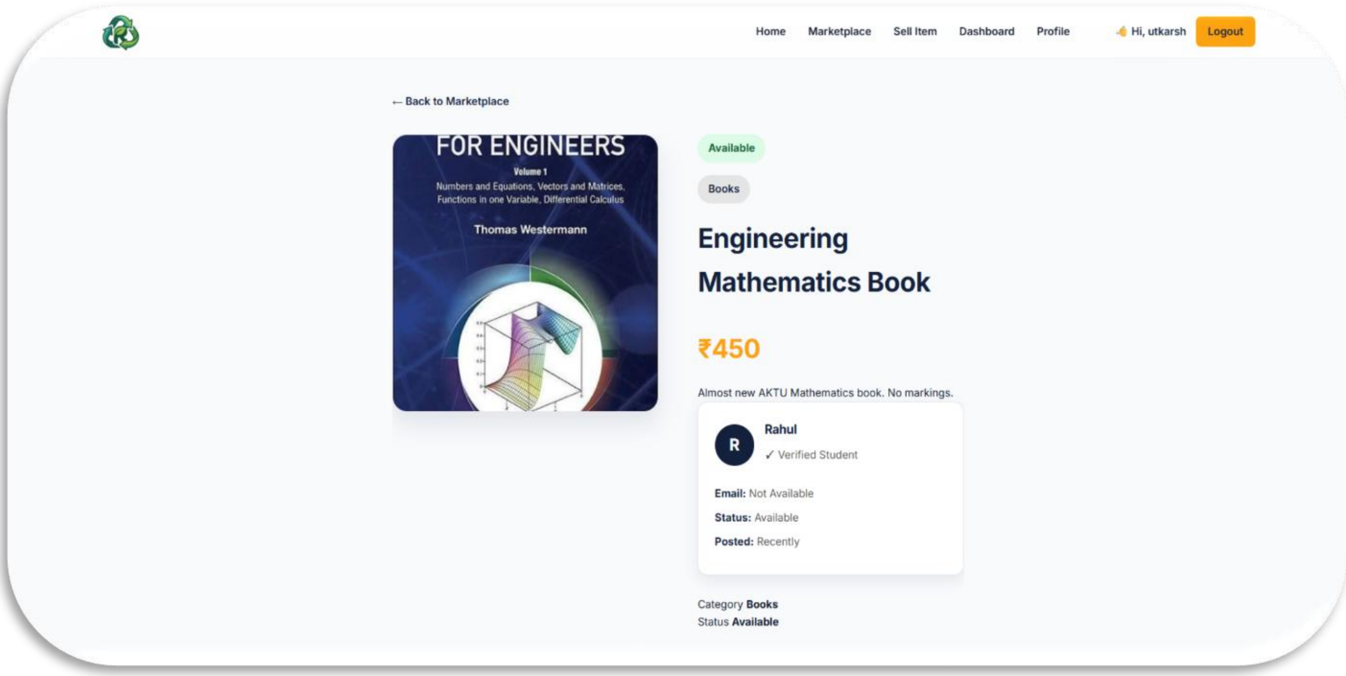
A.4 Dashboard



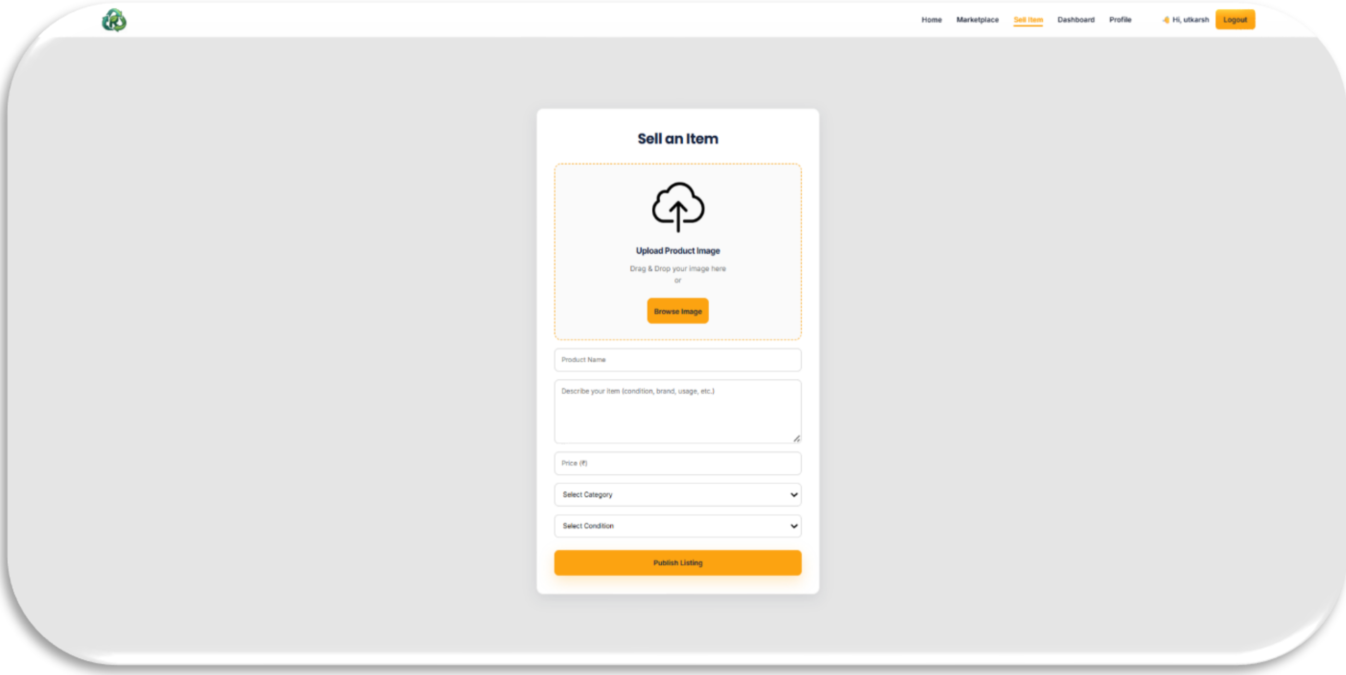
A.5 Marketplace



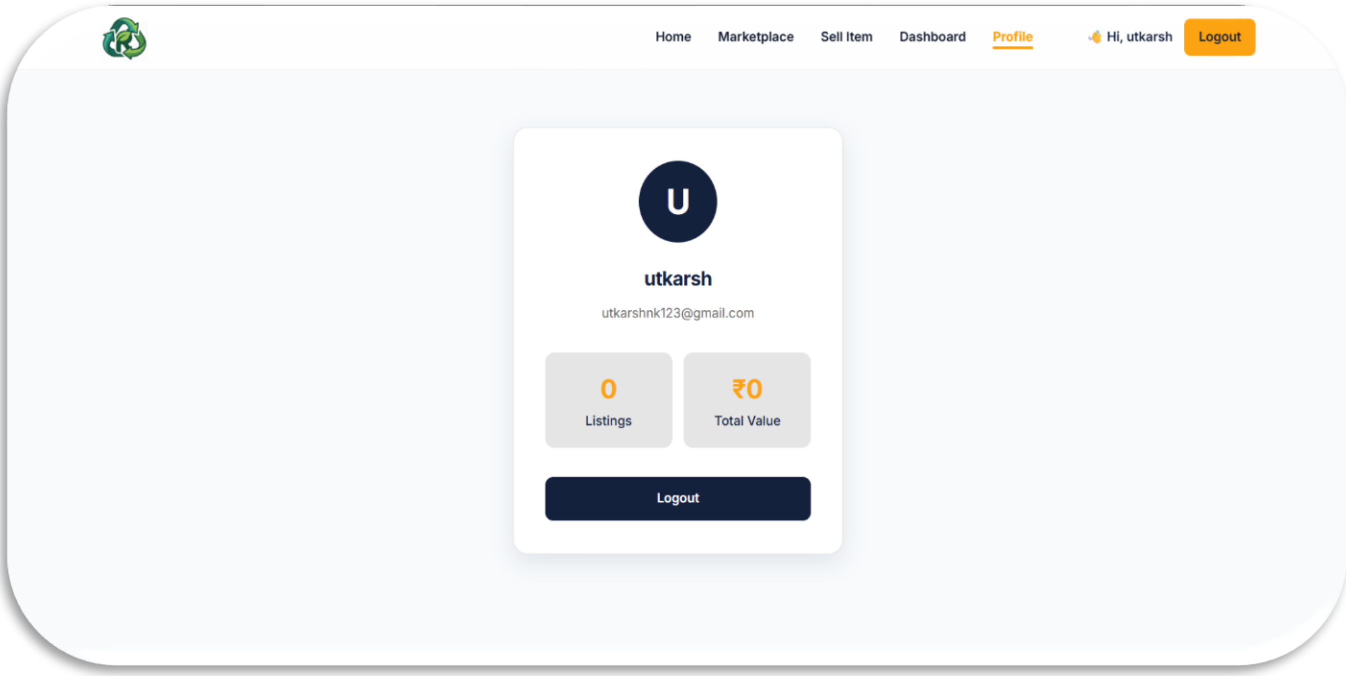
A.6 Product Details



A.7 Sell Product



A.8 User Profile



Appendix B: Project Structure

.git	06-Aug-26 1:04 AM	File folder	
client	05-Aug-26 10:08 PM	File folder	
pass	06-Aug-26 12:07 AM	File folder	
server	06-Aug-26 12:27 AM	File folder	
.gitignore	06-Aug-26 12:19 AM	Git Ignore Source File	17 bytes
README.md	04-Aug-26 10:18 PM	Markdown Source File	4.83 KB
ReNest.pptx	06-Aug-26 10:26 PM	Microsoft PowerPoint Pr...	19.0 MB
SummerInternshipReport.docx	07-Aug-26 1:01 AM	Microsoft Word Docum...	1.61 MB

Appendix C: GitHub Repository

Repository Name: ReNest

Repository URL:

https://github.com/Utkarsh0k/ReNest

Appendix F: Software Used

Software	Version
Windows 11	24H2
Visual Studio Code	Latest
Node.js	LTS
Express.js	5.x
MongoDB	Latest
Git	Latest
Postman	Latest
Google Chrome	Latest